

Methodology

Intended audience:

- Engineering university students familiar with Physics 1 content (kinematics and kinetics) and fundamental electrical concepts.

Motivation for improvements

With the rather low power ceiling imposed on teams for the 2026 and 2027 ruleset of the SAE Aero Design Regular Class competition, ensuring we get the maximum efficiency energy transfer is crucial in designing a competitive aircraft that can complete 1 lap of the airfield. To ensure we select optimal ESC, motor, and propeller combination we must measure the power transfer as it converts from electrical in the battery, to mechanical in the motor, and finally to propulsive in the propeller. In each transfer however, some power loss is to be expected due to internal resistance or drag. Measuring each type of power will allow us to determine how efficient each piece is at converting the power.

$$P_{\text{mechanical}} = \eta_1 P_{\text{electrical}}$$

$$P_{\text{propulsive}} = \eta_2 P_{\text{mechanical}}$$

$$\Rightarrow P_{\text{propulsive}} = P_{\text{electrical}}(\eta_1 \eta_2)$$

For the purpose of the improvements outlined in this document, assume propulsive power is already being measured with a load cell and electrical power is being recorded. Also, η_2 will vary depending on whether static or dynamic thrust is being measured.

Energy maximum

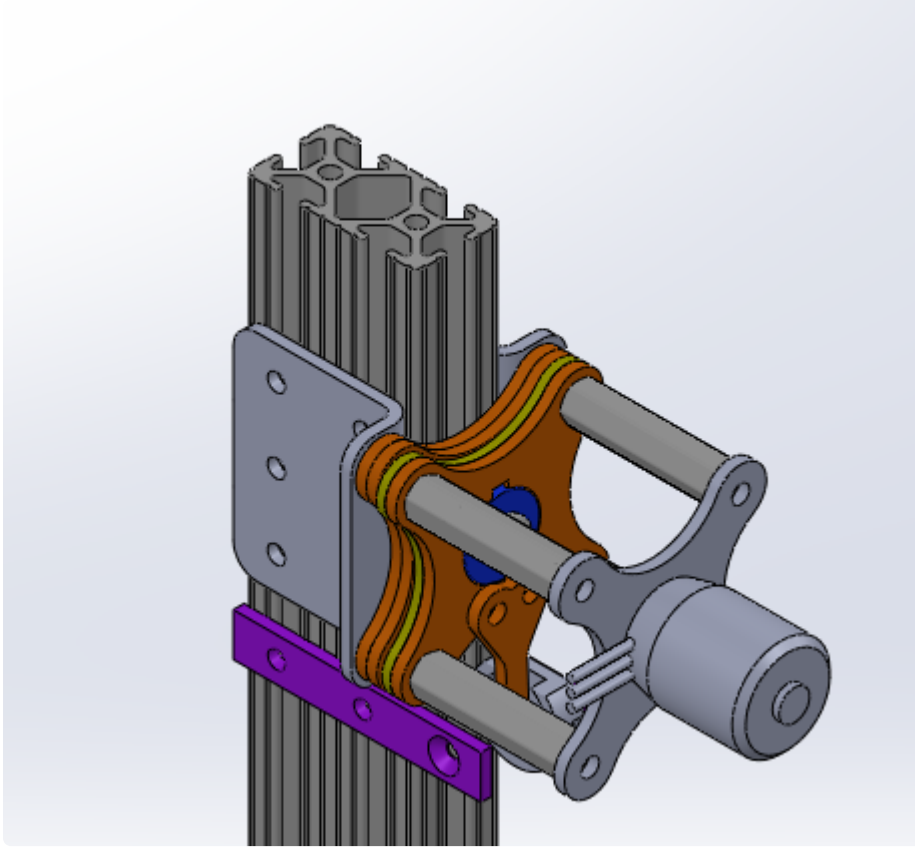
- 4S LiPo Battery (14.8 V nominal)
- 2.2 Ah capacity

First-Principle approach

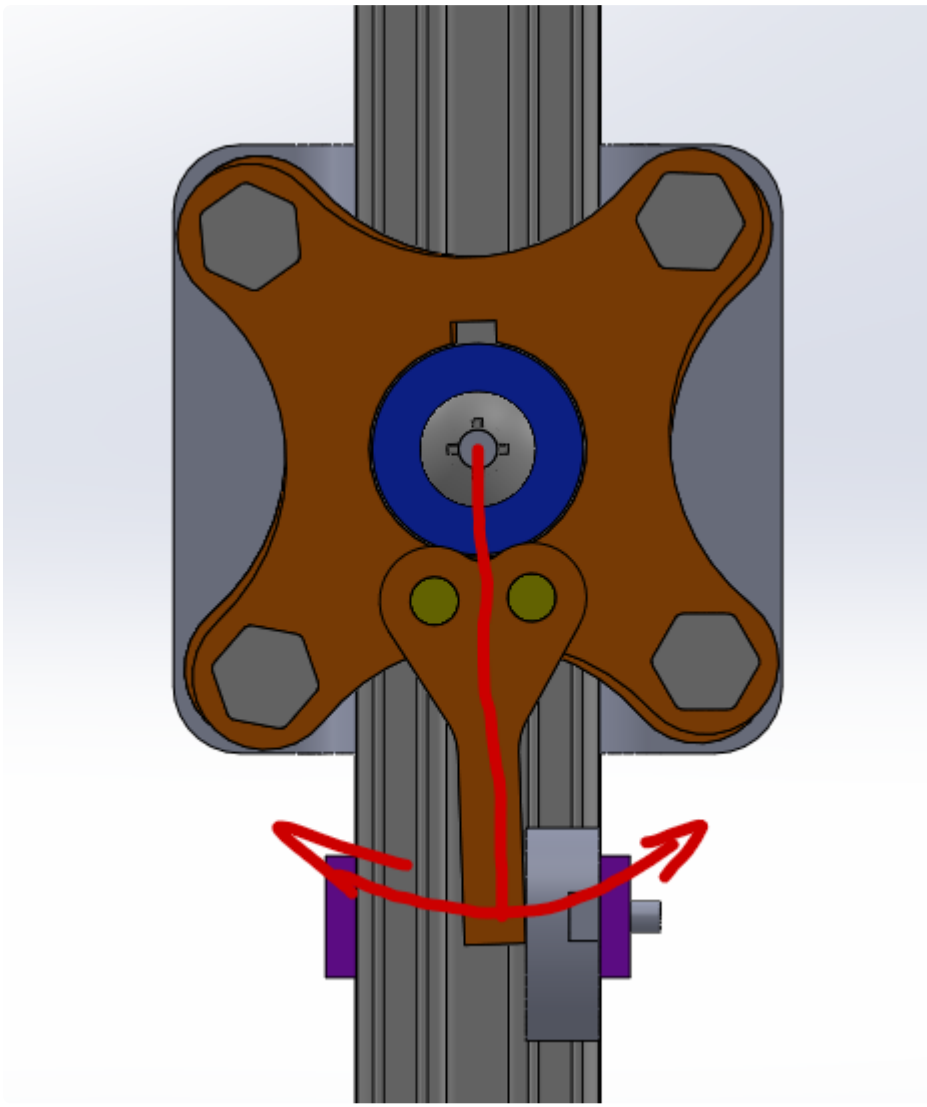
To measure torque generated by the motor, the [initial prototype](#) shown below exploits Newton's 3rd law of motion as it relates to rotational forces.

Newton's 3rd law of motion states that for every action there is an equal and opposite reaction. When the motor applies a torque to the propeller to spin it, the motor experiences an equal torque in the opposite direction to that of the propeller spin. If we allow the motor to have 1 degree of freedom about the motor's axis via a mount and attach an arm to that mount which applies a force to a load cell placed at a known distance away from the motor's axis, we can measure the torque imparted on the propeller from the motor by proxy at steady speed.

Initial prototype



In an effort to save money on these upgrades, many of the parts are designed to integrate with existing hardware. Everything in grey in the image above represents existing hardware in the club's possession while everything in color is new to the test stand. As shown in the image below, we allow the orange plate directly attached to the standoffs freely rotate about the motor's rotational axis. This orange plate also has an orange arm attached to it with the other end of the arm pressing into a load cell. The radius from the axis of rotation to the end of the arm is approximately 2.03 inches.



Expected Results

Expected data from this test stand should fit a linear approximation when mapping torque as a function of applied force with a slope equivalent to the radius.

$$\tau(F) = rF \sim f(x) = mx$$

Pairing this data with data gathered from a tachometer, we can calculate mechanical power generated by the motor.

$$P_{\text{mechanical}} = \tau\omega = \tau \frac{2\pi \text{RPM}}{60}$$

Pairing this new data of the mechanical power generated by the motor with electrical power, we can determine the efficiency of the motor and ESC.

$$\frac{P_{\text{mechanical}}}{P_{\text{electrical}}} = \eta_{\text{motor}}$$